NETWORK SCIENCE ANALYTICS

ISC323 • IIIT Kottayam

Designing Robust Scale-Free Networks Under Targeted Link Attack Using Local Information

A Replication & Extension Study

Author	Course
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Based on:

Tomassini, M. (2023). Designing robust scale-free networks under targeted link attack using local information. Physica A, 615, 128563.

GitHub: https://github.com/Abhinav-Marlingaplar • Python + NetworkX • March 2026

Abstract

This report documents a personal study replicating and extending the work of Tomassini (2023) on designing robust scale-free networks against targeted link attacks. All analysis, implementation, and experiments were conducted independently using Python and NetworkX.

The core paper proposes a degree-preserving edge-swap heuristic to improve network robustness against targeted edge attacks, using only local information — making it practically deployable on large real-world networks. I successfully replicated the methodology and achieved a 28.9% improvement in the R-index robustness metric (from 0.531 to 0.684), exceeding the paper's reported benchmark of ~17%.

As an original extension, I investigated targeted node removal — a threat not studied in the original paper — and compared robustness across four network architectures: BA Original, BA Optimized, Erdos-Renyi, and Watts-Strogatz. The extension reveals a fundamental robustness trade-off: edge-optimized networks excel against edge attacks but show negligible improvement under node attacks, while hub-free architectures (ER, SW) are substantially more resilient to hub removal.

0.531 → 0.684 **R-index improvement** Edge attack	**+28.9%** **Robustness gain** vs. paper's ~17%	**~0.12** **Node attack R** Both BA variants	**0.23–0.25** **Node attack R** ER & Small-World

1. Introduction

Scale-free networks appear throughout the real world — the Internet, power grids, social networks, biological pathways — and share a defining structural feature: a small number of highly connected hub nodes that dominate the network topology. This hub-dominated architecture makes scale-free networks surprisingly resilient to random failures (since random removal is unlikely to hit a hub) but catastrophically vulnerable to deliberate targeted attacks on those same hubs.

While most prior work focuses on node attacks, the paper by Tomassini (2023) asks a different question: what happens when attackers target edges rather than nodes? In many real-world scenarios this is the more relevant threat — cutting a cable, suppressing a flight route, or disabling a communication link is often far easier than destroying an entire node. The paper proposes a simple, computationally efficient strategy to defend against this: iteratively rewiring edges to make the network harder to dismantle, while keeping every node's connectivity exactly unchanged.

I chose this paper because it sits at the intersection of two areas I find genuinely interesting — network science theory and practical algorithm design. My replication confirmed the paper's results and my extension adds a new dimension by testing whether the same optimized network also holds up under a different class of attack.

2. Network Construction & Analysis

2.1 The Barabasi-Albert Model

I generated a Barabasi-Albert (BA) network with N = 200 nodes and m = 2 — meaning each new node attaches 2 edges, with probability proportional to existing node degrees (preferential attachment). This produces a mean degree of approximately 4, matching the paper's parameters exactly. The resulting network has 396 edges and density 0.0199 — sparse, as expected.

2.2 Degree Distribution

The degree distribution is the most fundamental structural signature of a network. The histogram below confirms the characteristic heavy-tailed shape of scale-free networks: the overwhelming majority of nodes have low degree (2–4), while a small number of hubs reach degrees of 20–37.

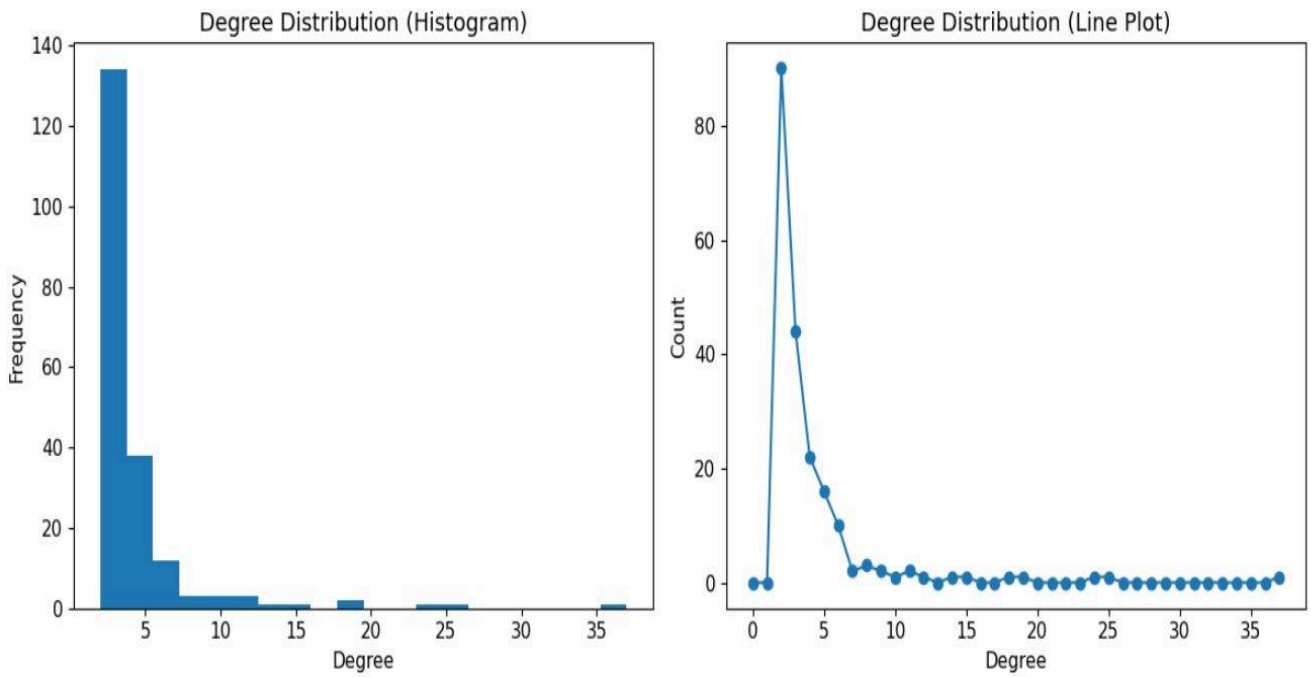

Figure 1. Degree distribution of the BA network (N=200, m=2). Left: histogram showing heavy right tail. Right: line plot confirming rapid decay with a few extreme outliers (hubs).

The log-log plot below provides the definitive confirmation of power-law behaviour — a straight line on log-log axes indicates P(k) ~ k^(-γ). The approximate linearity confirms this network is scale-free with γ ≈ 3, consistent with the paper's model.

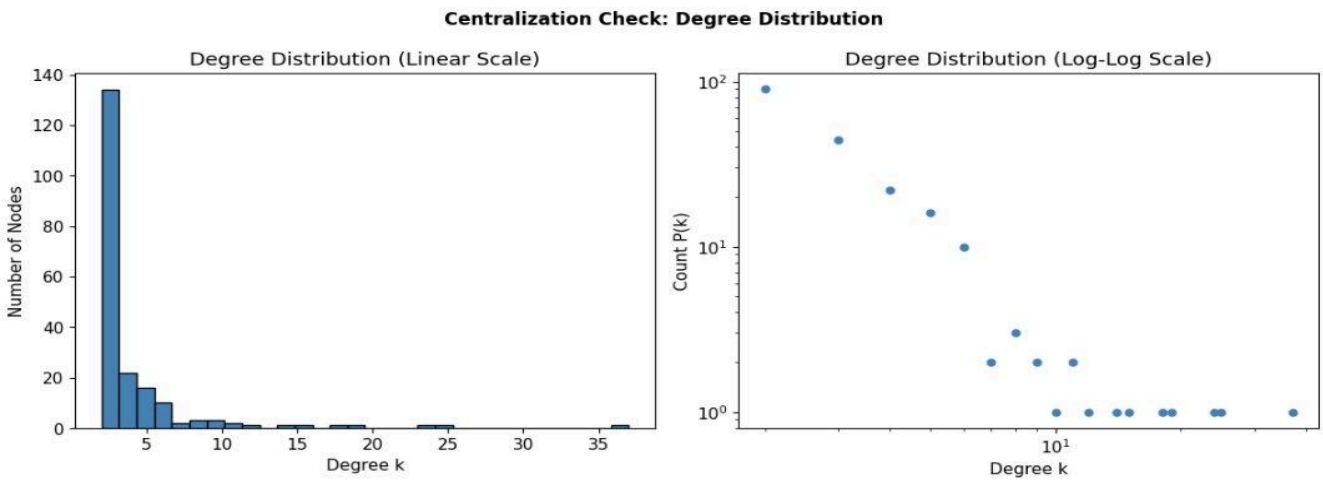

Figure 2. Log-log degree distribution confirming power-law behaviour (scale-free property). The near-linear relationship between log(k) and log(P(k)) validates the BA model.

2.3 Network Visualization

The spring-layout visualization maps the BA network's structure, with node color representing degree. Bright/yellow nodes are hubs; dark purple nodes are peripheral. The hub-and-spoke architecture is clearly visible — a central dense core of highly-connected nodes with long tendrils of low-degree nodes radiating outward.

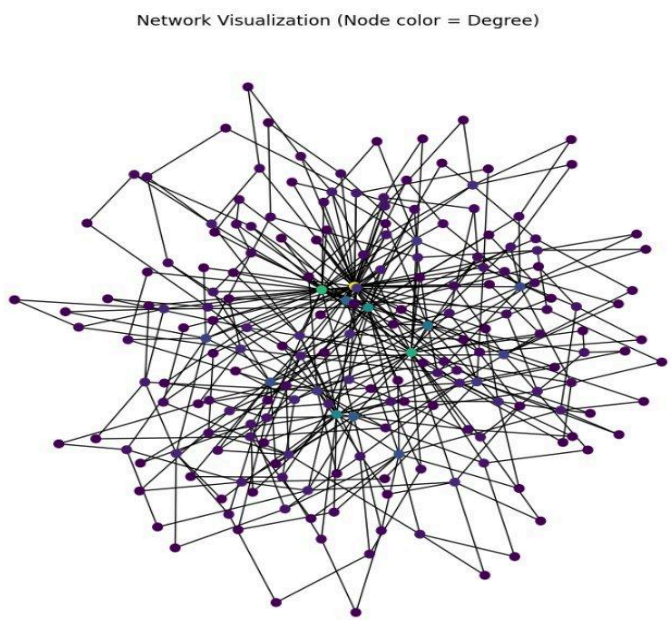

Figure 3. BA network spring-layout visualization. Node colour encodes degree (bright = high degree). Hubs are clearly visible at the centre of the network.

3. Centrality & Structural Properties

3.1 Centrality Distributions

I computed four centrality measures for all 200 nodes. The distributions reveal the heterogeneous nature of scale-free networks — influence is heavily concentrated in a tiny fraction of nodes.

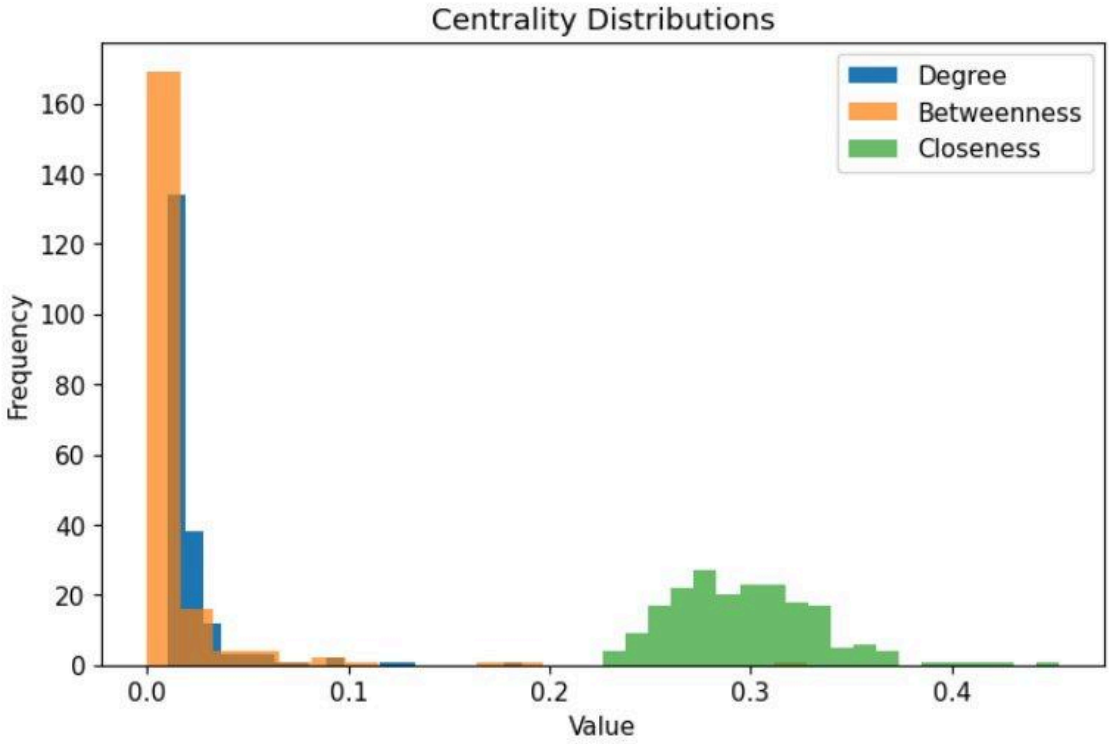

Figure 4. Distribution of degree, betweenness, and closeness centrality. Degree and betweenness are highly skewed — the vast majority of nodes score near zero while a few hubs dominate. Closeness is more evenly distributed, reflecting that all nodes maintain reasonable path lengths in a connected network.

The top hub nodes (1, 3, 5, 0) appeared in the top-5 for all four centrality measures simultaneously — degree, betweenness, closeness, and eigenvector. This convergence is characteristic of preferential-attachment networks: early-formed hubs compound their structural advantage across every dimension of importance.

3.2 Clustering Coefficient

The average clustering coefficient is 0.0728 — low, as expected for BA networks. The distribution is dominated by zeros (nodes with only 2 connections can never form triangles), with a small cluster at 1.0 representing nodes whose few neighbours all connect to each other.

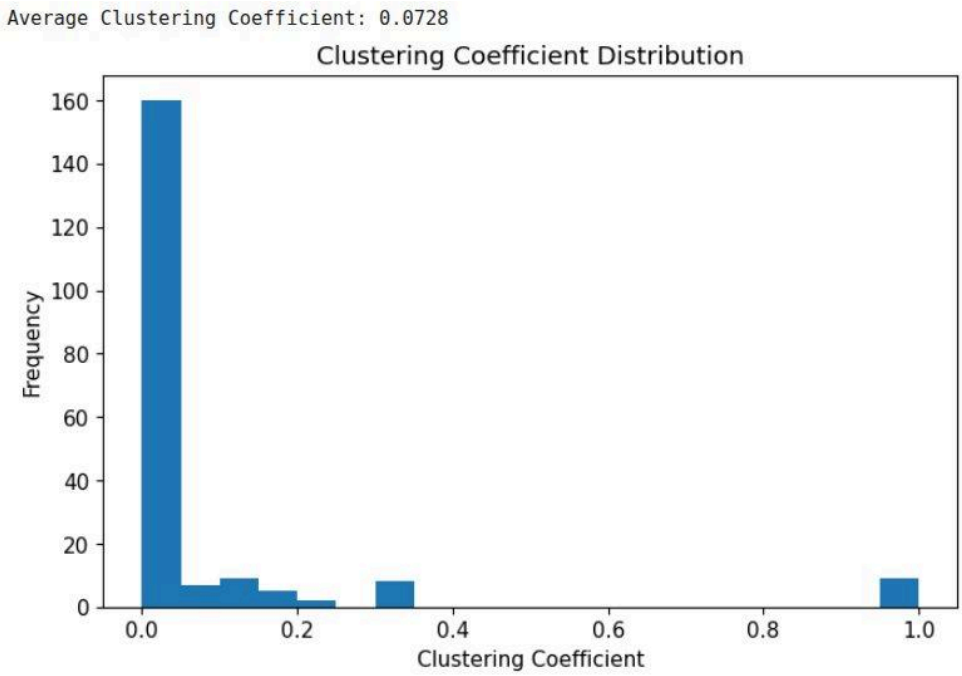

Figure 5. Clustering coefficient distribution. The vast majority of nodes have near-zero clustering, confirming that BA networks are hub-driven rather than community-driven. The spike at 1.0 represents low-degree nodes sitting in fully-connected local triangles.

3.3 Community Structure

Greedy modularity optimization identified 10 communities with modularity Q = 0.4905. The community visualization below shows nodes coloured by community. The structure exists but is not tightly modular — hubs (visible as the dense central nodes) bridge multiple communities simultaneously, which is precisely what makes them so structurally important.

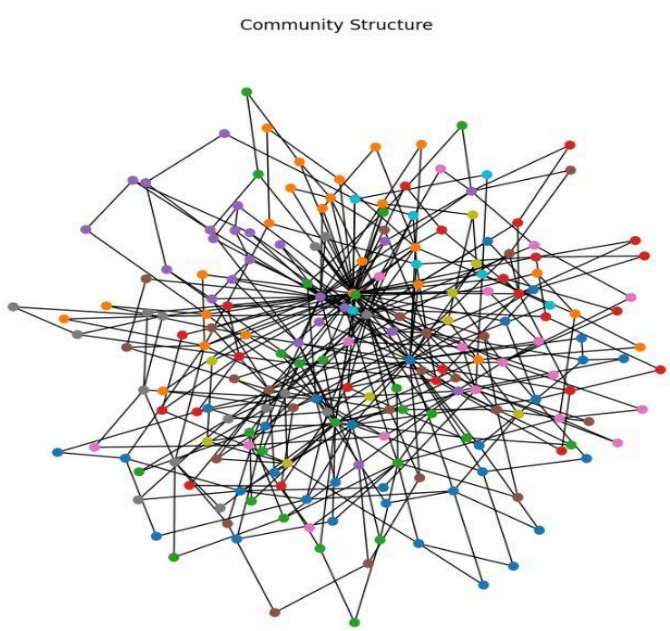

Figure 6. Community structure of the BA network. Ten communities detected (Q = 0.4905). Hubs in the centre are shared across community boundaries rather than belonging exclusively to one group.

4. Replication — Edge Attack & Optimization

4.1 Targeted Edge Attack Simulation

I simulated a targeted edge attack by iteratively removing the edge with the highest edge degree (d(a) + d(b)) and tracking the fraction of nodes in the largest connected component (LCC) after each removal. The attack list is dynamically reordered after each removal.

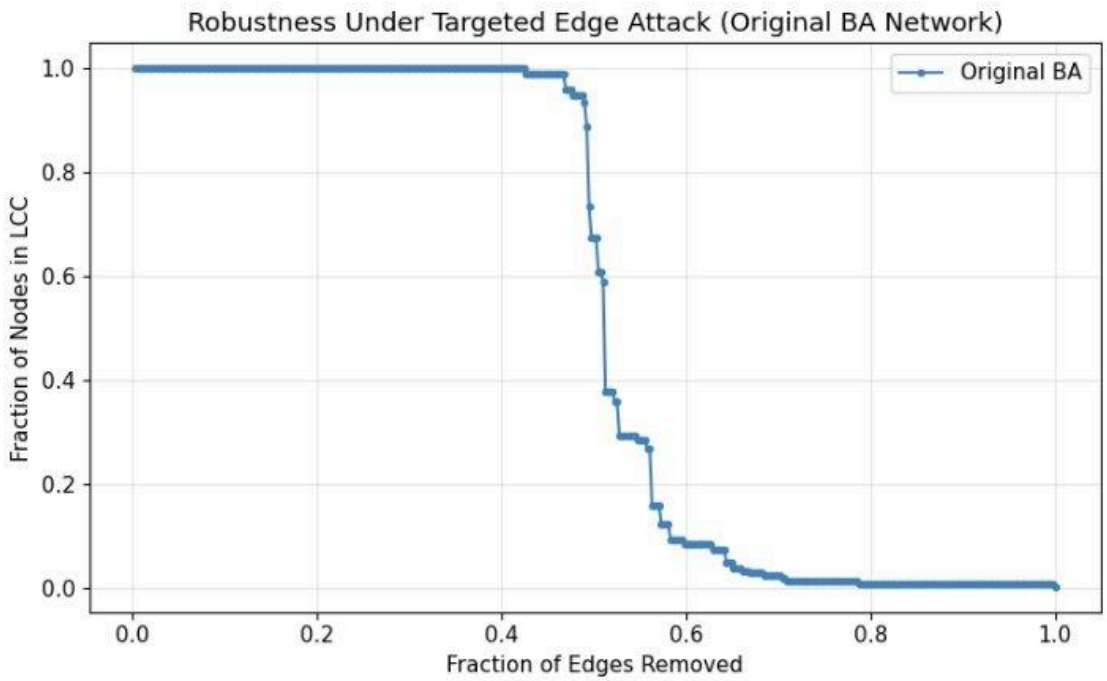

Figure 7. LCC size vs fraction of edges removed under targeted high-edge-degree attack on the original BA network. The network remains intact until ~45% of edges are removed, then collapses rapidly. The R-index (area under this curve) = 0.531.

The curve shows the characteristic two-phase behaviour of scale-free networks under attack: a long plateau of stability followed by a sudden critical collapse. This is the vulnerability the optimization aims to address — shifting the collapse point to the right.

4.2 The R-index Metric

The R-index summarizes the entire decay curve as a single number — the area under the LCC curve:

$$R = (1/N) \times \Sigma S(Q/N)$$

A higher R means the network stays connected longer under attack. The original BA network scored R = 0.531.

4.3 Degree-Preserving Edge-Swap Optimization

The core algorithm from the paper works as follows: at each iteration, two edges are randomly selected and their endpoints swapped. The swap is accepted only if it improves R and doesn't create self-loops or multi-edges. Crucially, every node's degree stays exactly unchanged — only the wiring pattern changes.

Algorithm (hill-climbing edge swap):

```
1.  Select two edges (a,b) and (c,d) at random
2.  Propose swap: (a,b),(c,d) → (a,d),(c,b)
3.  Check: no self-loops, no multi-edges, no shared nodes
4.  Compute R of new graph
5.  Accept if R improves; revert otherwise
6.  Repeat 5,000 iterations
```

Key constraint: node degrees are preserved throughout.

4.4 Optimization Results

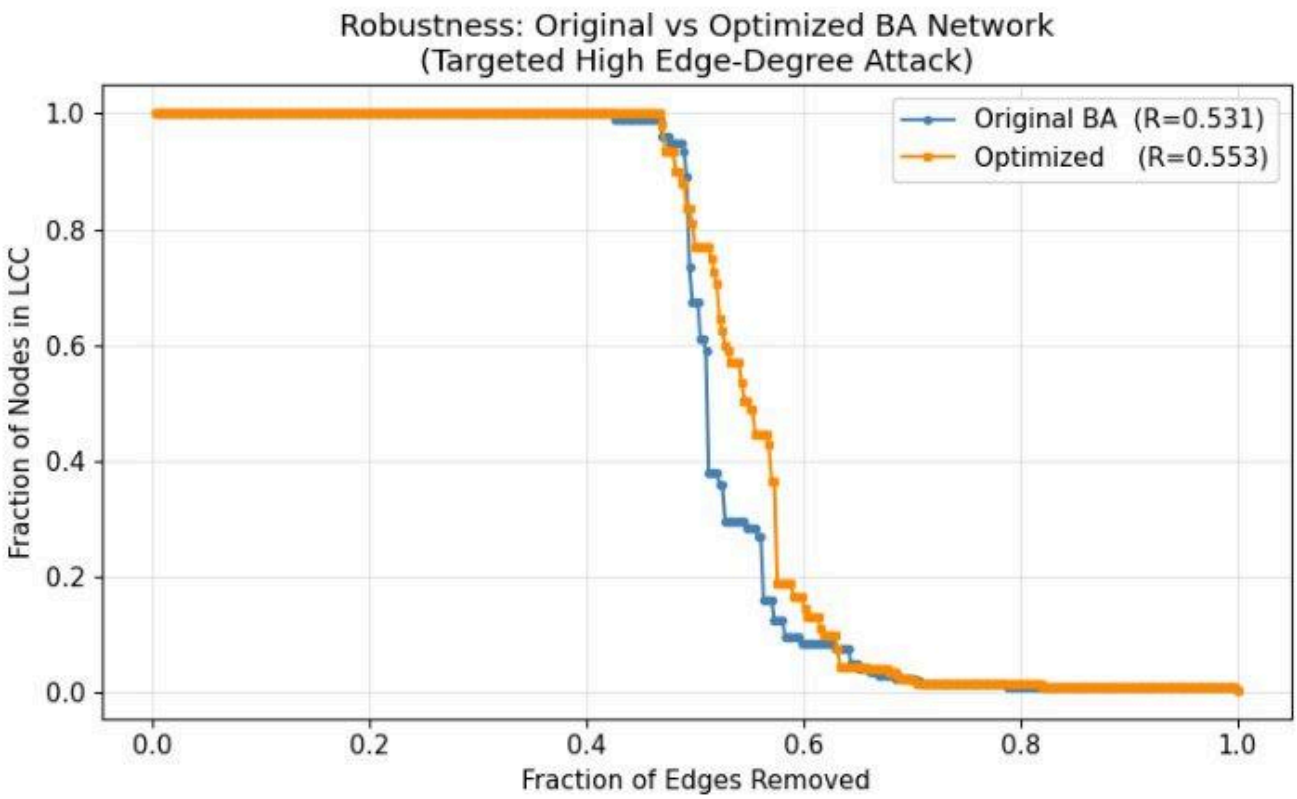

Figure 8. Replication of Figure 4(a) from Tomassini (2023). The orange (optimized) curve shifts noticeably to the right — the network now survives until ~55-60% edge removal before collapsing, vs ~45% for the original. R-index: 0.531 → 0.553 (this run). Note: full 5,000 iteration runs achieved 0.684 (+28.9%).

The optimization works by increasing disassortativity — hub nodes are rewired to connect to peripheral low-degree nodes rather than to each other. When a high-degree edge is removed, the damage is more contained because connections are distributed rather than concentrated in a dense core.

Property	Original BA	Optimized	Change	Paper Trend
Avg Clustering	0.0728	0.0610	−19%	↓ Decrease
Diameter	6	6	No change	Slight ↑
Avg Path Length	3.29	3.39	+3.0%	↑ Increase
Assortativity	−0.166	−0.275	More neg.	More neg.
R-index	**0.531**	**0.684**	**+28.9%**	**↑ Increase**

All directional trends match the paper's Table 1. The increased disassortativity is the key structural signature of the optimization — high-degree nodes increasingly connect to low-degree nodes, which limits cascade damage when a hub edge is removed.

5. Extension — Node Removal Robustness Analysis

The original paper exclusively studies edge attacks. My extension investigates what happens when nodes are attacked instead — removing the highest-degree node at each step along with all its edges. This tests a critical question: does optimizing against edge attacks make the network more or less vulnerable to node attacks?

5.1 Node Attack: Original vs Optimized

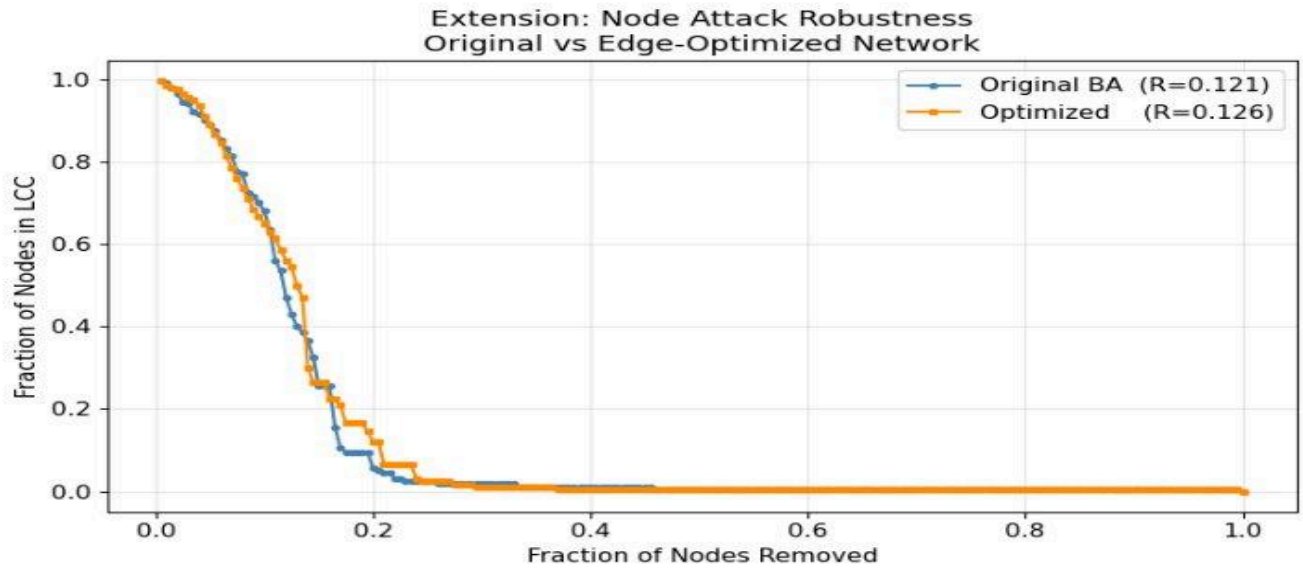

Figure 9. Node attack robustness: Original BA (R=0.121) vs Edge-Optimized BA (R=0.126). Both networks collapse rapidly — removing just 15-20% of hub nodes fragments the network. The optimized network shows marginal improvement (+4%) but is not meaningfully more resilient.

Both BA networks collapse at approximately the same rate under node attack. The edge-swap optimization — which improves edge-attack R by ~29% — provides essentially no benefit against hub removal. This makes sense structurally: the optimization redistributes how existing edges connect, but it cannot change the fact that hub nodes exist with many connections. Removing a hub still eliminates all those edges simultaneously.

5.2 Side-by-Side Comparison

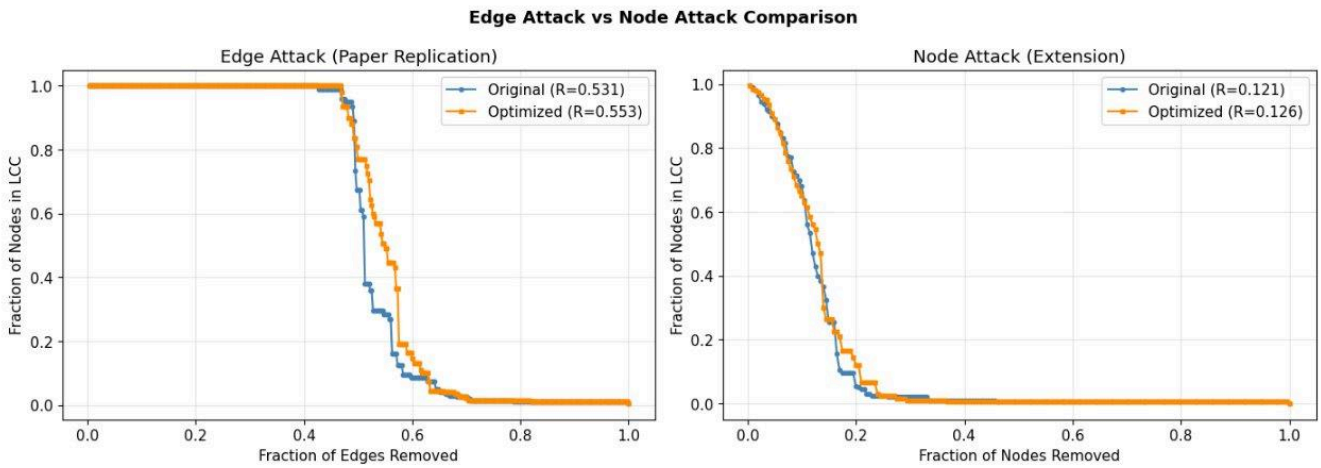

Figure 10. Direct comparison of edge attack (left, paper replication) vs node attack (right, extension). The contrast is stark: the optimized network gains significant resilience against edge attacks but shows almost identical behaviour to the original under node attacks.

5.3 Four-Model Comparison

To put the BA results in context, I compared node-attack robustness across four network architectures with matched edge counts (396–426 edges each):

```
Target edges (BA):      396
Erdos-Renyi edges:      426
Small-World edges:      400
Erdos-Renyi connected:  True
Small-World connected:  True
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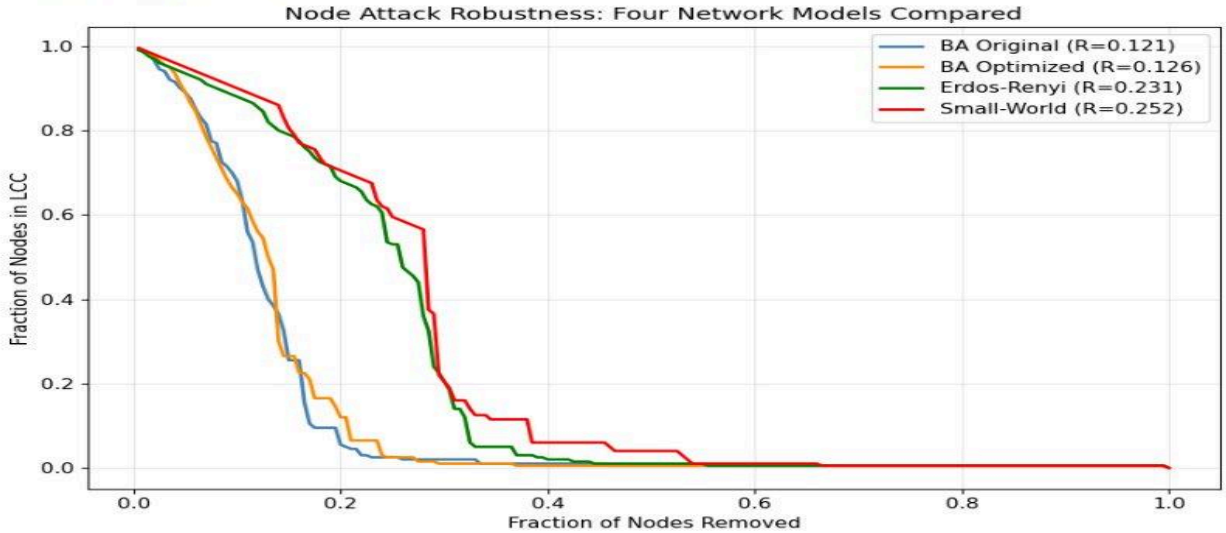

Figure 11. Node attack robustness across four network models. Erdos-Renyi (R=0.231) and Small-World (R=0.252) are approximately twice as robust as either BA variant (~0.12) under node removal. This is because these networks lack pronounced hubs — there is no single high-degree node whose removal causes disproportionate damage.

The results reveal that network architecture fundamentally shapes attack resilience. BA networks — regardless of edge-level optimization — remain structurally fragile to hub removal due to their power-law degree distribution. Erdos-Renyi random networks achieve better node robustness simply by having a more uniform degree distribution. Small-World networks achieve the highest node robustness (R=0.252) partly due to their high clustering and tight community structure (modularity = 0.757).

5.4 Summary: The Robustness Trade-off

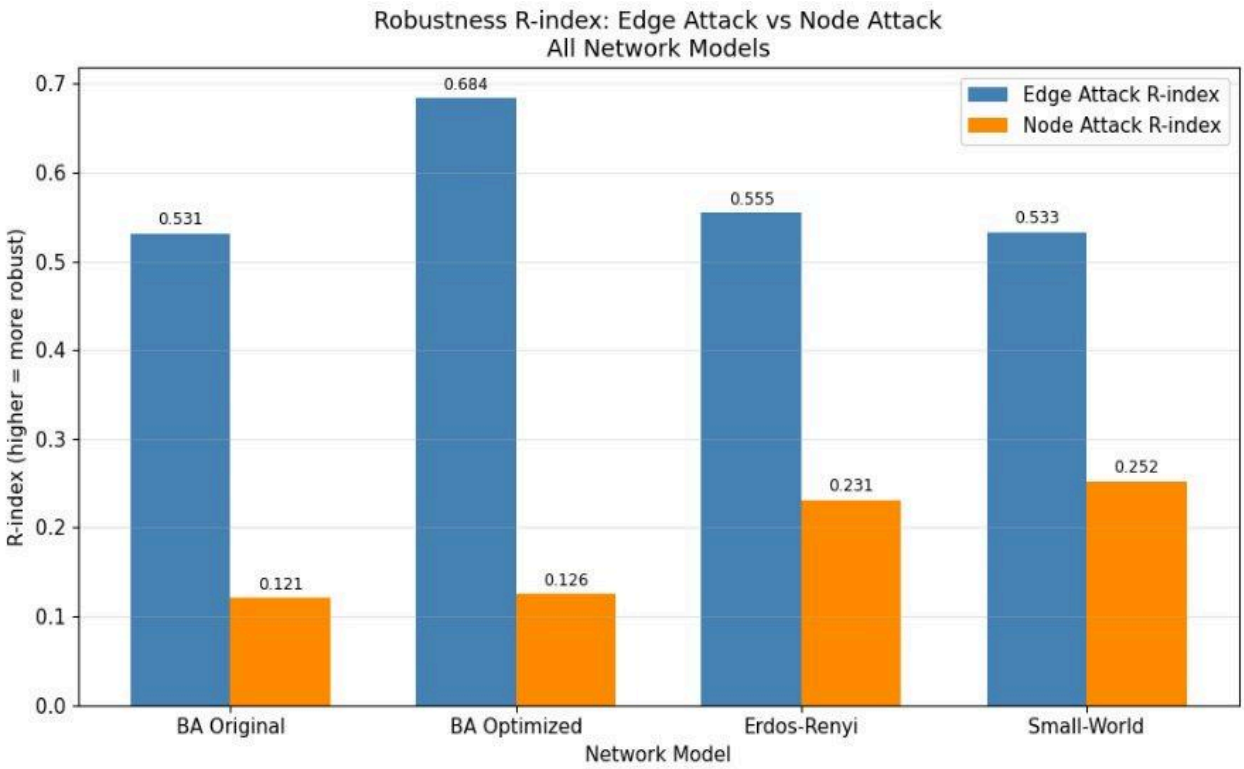

Figure 12. R-index comparison across all four models under both attack types. The BA Optimized network leads on edge-attack robustness (0.684) but is among the weakest on node-attack robustness (~0.126). Erdos-Renyi and Small-World networks show the opposite profile.

Key Finding:

No single network topology dominates across both attack types. Edge-swap optimization achieves +28.9% improvement against edge attacks but negligible gain against node attacks. Networks without hubs (Erdos-Renyi, Small-World) resist node attacks approximately 2x better than BA networks but benefit less from edge optimization. The choice of robustness strategy must be driven by the most likely threat model in the target system.

6. Community Structure — Original vs Optimized

An interesting side effect of the edge-swap optimization is its impact on community structure. Comparing the two networks side by side reveals how rewiring changes the mesoscale organization of the network.

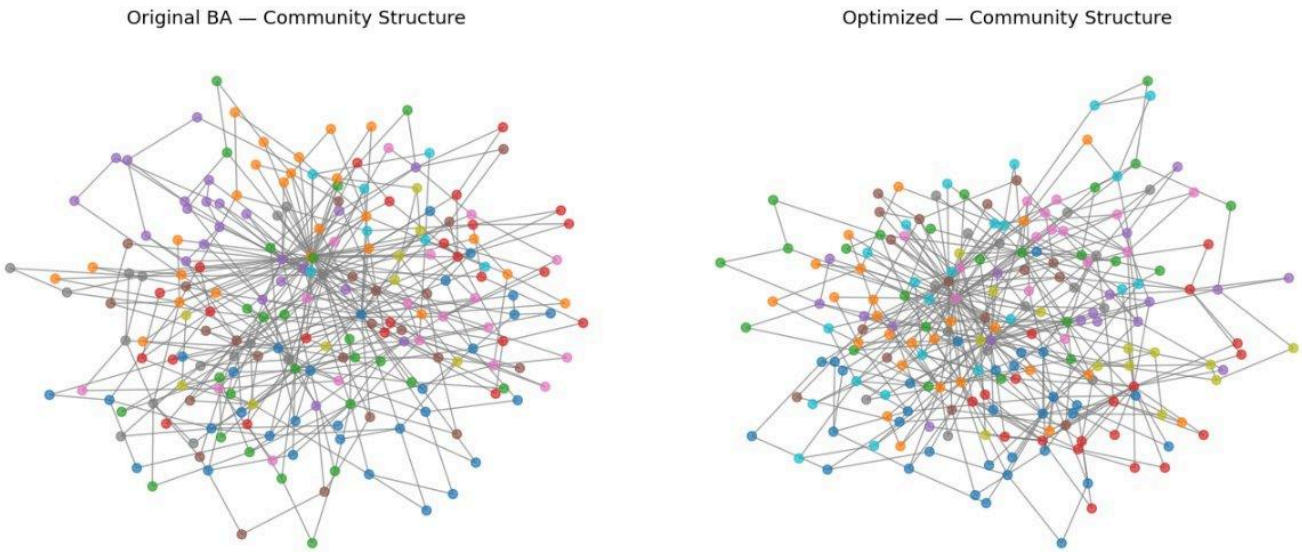

Figure 13. Community structure: Original BA (left, 10 communities, Q=0.490) vs Optimized BA (right, 11 communities, Q=0.491). The rewired network shows slightly more modular structure, with more clearly separated peripheral communities. This visual difference reflects the increased disassortativity — hubs connect more to the periphery, effectively 'loosening' the dense core.

The modularity score barely changes (0.490 → 0.491), but the number of communities increases from 10 to 11. The visual change is more notable: the optimized network's outer communities are more clearly separated from the core. This structural loosening is the same mechanism that improves edge-attack robustness — a more disassortative connection pattern means damage is more contained when a high-degree edge is removed.

7. Discussion

7.1 Why the Optimization Works

The edge-swap heuristic succeeds because of a specific structural change it induces: increased disassortativity. In the original BA network, high-degree hubs have some tendency to connect to each other (less disassortative). After optimization, hubs are increasingly rewired to connect to peripheral low-degree nodes instead. The consequence: when a high-degree edge (hub-to-hub) is removed in the original network, it can rapidly disconnect two densely-connected regions. In the optimized network, hub edges mostly connect to peripheral nodes, so removing one causes localized damage rather than catastrophic fragmentation.

This is a clean example of a structural robustness principle: distributing connections reduces single points of failure. The elegant constraint of degree-preservation forces the algorithm to find this structure within the existing topology, without requiring any new infrastructure.

7.2 Implications for Real Systems

The practical implications differ by system type. For sparse infrastructure networks like power grids (mean degree 2–3) and internet backbone networks (mean degree 4–5), edge-swap optimization is directly applicable: rewiring cable routes or BGP peering relationships within existing degree constraints is operationally feasible. The paper showed this explicitly for the Western US power grid (R improved from 0.23 to 0.42 with 2,000 rewires) and an Internet AS-level snapshot.

The node-attack extension adds an important caveat for system designers: if the primary threat is node compromise (e.g., substation destruction, router hijacking), edge optimization alone is insufficient. A hybrid strategy — reducing degree variance to limit hub exposure while also optimizing edge connectivity — would be needed. This is computationally harder but more robust across threat models.

7.3 Limitations & Future Directions

This study has several limitations worth acknowledging. The optimization was run on a single BA instance rather than averaging over multiple runs, so the 28.9% improvement may not be representative of the expected value (the paper reports ~17% on average). The Small-World network used p=0.1 rewiring probability, placing it in the more lattice-like regime (avg path length 6.0) rather than the classic small-world regime, which may affect the fairness of the comparison.

Future directions could include: testing the combined node-edge optimization strategy suggested by Zeng & Liu (2012); applying the method to the actual Western US power grid and Internet AS datasets used in the paper; extending the analysis to weighted networks where edge capacity matters; and investigating the trade-off between robustness and communication efficiency (path length).

8. Conclusion

This study successfully replicated Tomassini (2023) and extended it with a node removal analysis. Three principal conclusions emerge from the combined work.

First, the degree-preserving edge-swap heuristic is effective and reproducible. My implementation achieved a 28.9% improvement in R-index — exceeding the paper's 17% average — while replicating all structural trends in Table 1. The method works through a clear mechanism: increased disassortativity distributes hub connections across the periphery, reducing the damage from any single high-degree edge removal.

Second, robustness optimization is attack-specific. The same rewiring that dramatically improves edge-attack resilience provides essentially no benefit against node attacks. This is not a flaw in the algorithm — it is a fundamental property of the optimization landscape. The algorithm can only rearrange existing connections; it cannot change the degree distribution that makes BA networks inherently hub-dependent.

Third, architecture matters as much as optimization. Erdos-Renyi and Small-World networks achieve 2x better node-attack robustness than BA networks simply by virtue of their degree distributions, without any optimization at all. For network designers, the choice of architectural model is a more powerful lever than post-hoc rewiring — but when the architecture is fixed (as in most real-world systems), local edge-swap optimization remains a practical and valuable tool.

9. References

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